7 Read the passage and answer the questions that follow.

World Cup footballs have different panel shapes, seam patterns and surface textures. These features affect the airflow around the ball and hence the forces acting on it during flight.

Table 7.1 shows the physical properties of three official FIFA World Cup footballs released between 2010 and 2026.

Table 7.1

Ball (Year)	Number of Panels	Total Seam Length / m	Seam Width / mm	Seam Depth / mm
Jabulani (2010)	8	1.98	2.2	0.5
Al Rihla (2022)	20	3.52	5.8	1.6
Trionda (2026)	4	2.50	5.1	1.3

Figure 7.1 shows the seam and a panel of the Trionda.

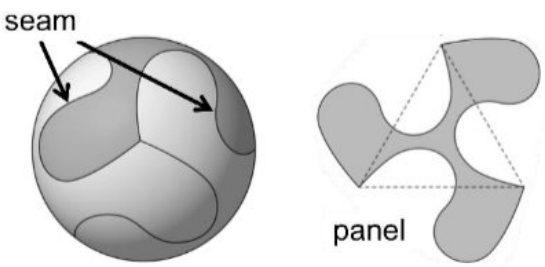


Figure 7.1

Modern FIFA World Cup footballs are engineered so that reducing the number of panels does not result in an excessively smooth surface, which could lead to undesirable changes in behaviour.

Although the Trionda has fewer panels than the Al Rihla, each panel contains three pronounced grooves that act as additional roughness elements. As a result, the Trionda has an overall effective surface roughness comparable to that of the Al Rihla. The two footballs therefore exhibit similar drag-crisis behaviour.

Wind-tunnel tests allow researchers to measure the drag coefficient C_D of a ball at different speeds.

The drag force F_D on a football is given by

$$F_D = \frac{1}{2} C_D \rho A v^2$$

where

ρ : density of air
A : cross-sectional area of the ball
v : speed of the ball relative to the air

World Cup footballs are designed to have a mass of 0.430 kg and a diameter of 0.220 m to satisfy FIFA regulations. The density of air is taken to be 1.20 kg m^{-3}.

Figure 7.2 shows the variation with the speed v of the drag coefficient C_D for each football.

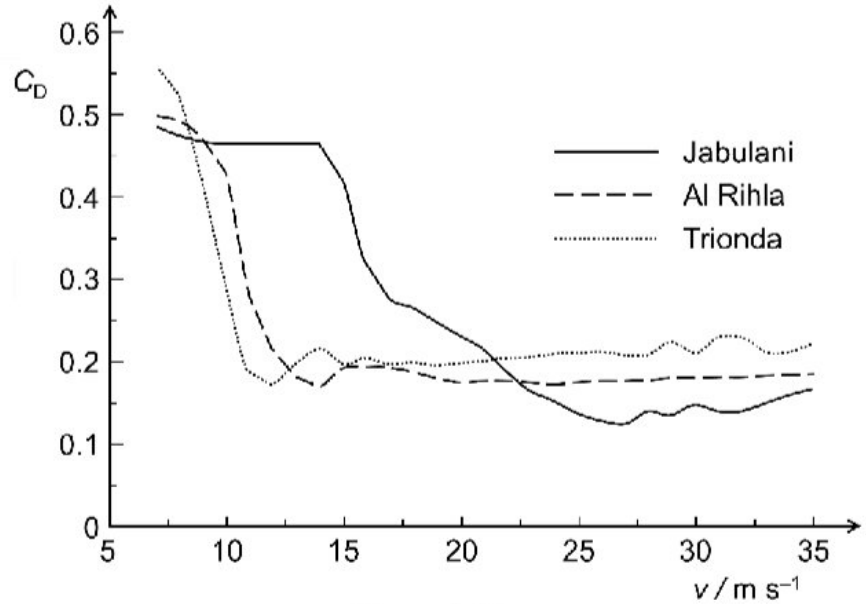


Figure 7.2

As the speed increases, the drag coefficient decreases rapidly over a range of speeds before becoming approximately constant. This behaviour is referred to as the drag-crisis region. The critical speed is defined as the speed at which the minimum drag coefficient occurs within the drag-crisis region. The critical speeds for the Trionda and Al Rihla are 11.9 m s^{-1} and 13.9 m s^{-1} respectively.

The flight of each football was simulated under identical launch conditions. Each football was kicked with the same initial speed, at the same launch angle of 25° and without spin. The predicted trajectories are shown in Figure 7.3.

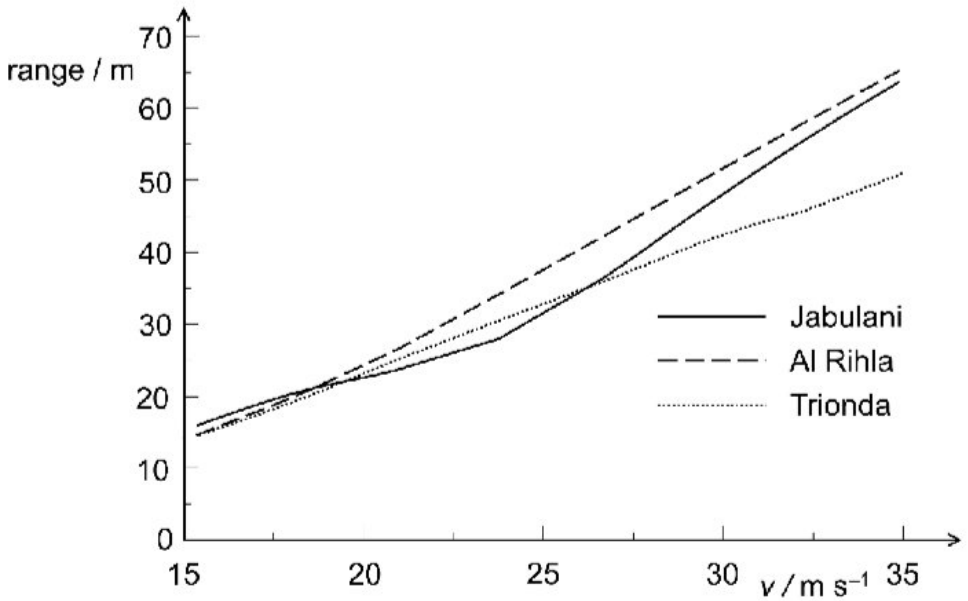


Figure 7.3

Match analysis indicates that short passes, used to maintain possession and accuracy between nearby teammates, are typically made at speeds between 12 and 15 m s^{-1}. Long kicks, used to move the ball over greater distances, are typically made at speeds exceeding 25 m s^{-1}.

(a) State the critical speed of the Jabulani.

critical speed = ... m s^{-1} [1]

(b) Using Table 7.1 and Figure 7.2, suggest why the Jabulani requires the highest speed to reach the drag-crisis region.

.....

.....

.....

.....

.....

..... [3]

(c) Determine the drag force acting on the Jabulani at 30 m s^{-1}.

drag force = ... N [2]

For Examiner's Use

(d) The designers of the Trionda aimed to reduce drag during short passes while limiting the distance of long kicks.

Explain how the data show that these aims have been achieved.

.....

.....

.....

.....

.....

..... [3]

(e) Both the Jabulani and Trionda travel at 30 m s^{-1}.

Determine the ratio

$$\frac{\text{rate of kinetic energy transferred to the surrounding air by the Jabulani}}{\text{rate of kinetic energy transferred to the surrounding air by the Trionda}}$$

ratio = ... [3]

For Examiner's Use

(f) The coefficient of ballistic performance B indicates how important aerodynamic forces are on the trajectory of a ball in flight. It is given by

$$B = \frac{mg}{\rho A v^2}$$

where m is the mass of the ball.

(i) Derive an expression for the ratio $\frac{F_D}{mg}$ in terms of C_D and B.

[1]

(ii) Determine the value of $\frac{F_D}{mg}$ for the Trionda at a speed of 30 m s^{-1}.

$\frac{F_D}{mg}$ = ... [1]

(iii) Hence, state and explain whether air resistance is negligible for the Trionda at this speed.

.....

..... [1]

(iv) State and explain the trajectory of the ball when B tends to a large value.

.....

.....

.....

..... [2]

(g) A spinning football also experiences an additional force known as the Magnus force. Figure 7.4 shows the airflow around a football spinning anticlockwise immediately after it leaves the player's foot and travels to the left.



Figure 7.4

Experiments show that regions where the airflow speed is higher correspond to regions where the air pressure is lower.

(i) Explain why the top of the ball is at a higher pressure than the bottom of the ball.

.....

.....

..... [1]

(ii) The Magnus force causes the spinning football to curve with its speed remaining approximately constant over a short time interval.

State and explain the work done by the Magnus force on the ball.

.....

.....

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..... [2]

[Total: 20]